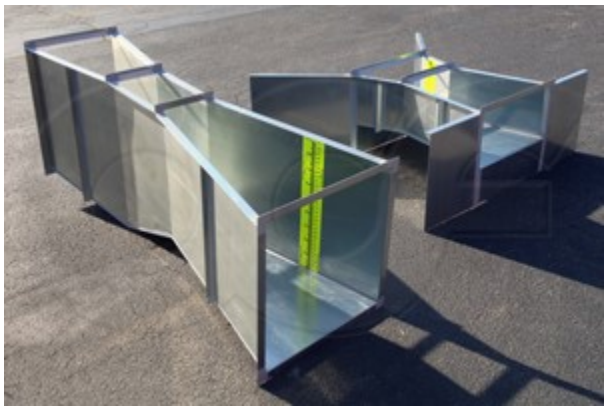
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Flumes for Accurate Flow Measurement

4-5 minutes



Flumes are specially shaped, engineered structures that are used to measure the flow of water in open channels. Flumes are static in nature - having no moving parts - and develop a relationship between the water level in the flume and the flow rate by restricting the flow of water in a variety of ways.

Flumes can accelerate slow, sub-critical (Fr<1) flow to a supercritical state (Fr>1) by:

- A change in elevation
- A contraction of the sidewalls
- Or a combination of the two

Accelerating slow flow to a supercritical state creates upstream conditions where, under free flow conditions, the flow rate can be determined by measuring the water level at a **single, defined point**

in the flume (H_a).

The relationship between the water level at the point of measurement (H_a) and the flow rate can be obtained by test data (short-throated flumes) or derived formula (long-throated flumes).

Flume Accuracy

When properly installed, flumes can be accurate to within +/-2-5% (for the flume itself) - with overall system accuracy for a typical installation being +/-10% (when all factors are considered). This accuracy is similar to that of weirs - although weirs are slightly more accurate under ideal conditions.

Flume Advantages

While similar in accuracy, flumes have distinct advantages over weirs, including:

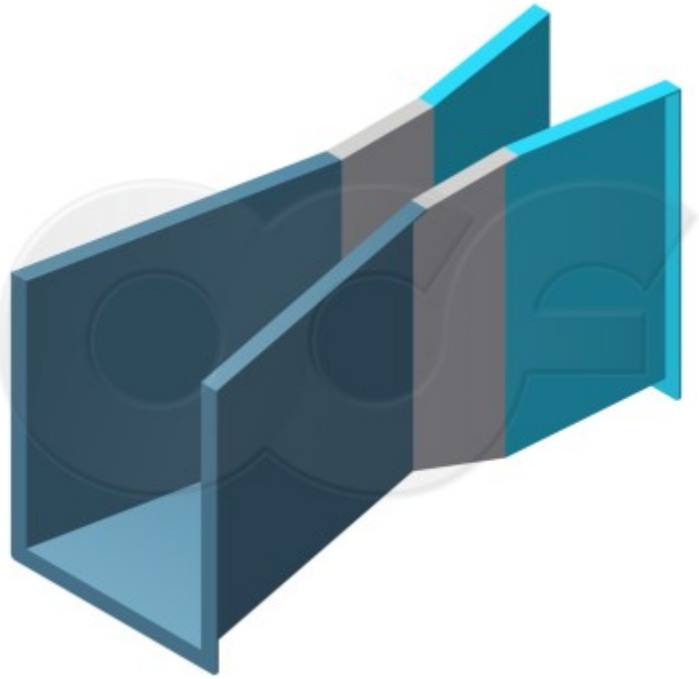
- The ability to measure higher flow rates than a comparably sized weir
- Less head loss (generally 1/4th that of a weir)
- The ability to pass debris more readily
- Wide range of styles and sizes
- Off-the-shelf availability
- Smaller installation footprint
- Less rigorous maintenance requirements

Also, most styles of flumes are resistant to changes or restrictions in the downstream hydraulics (submergence) - something not found with weirs.

Sections of a Flume

Most flume styles consist of three parts: a converging section (inlet), a throat, and a discharge section (outlet).

- Flow is restricted in the converging section (dark blue) and is accelerated as it passes into the throat. Flow in the converging section must enter at a sub-critical state [typically with a Froude number (F_r) no more than 0.5] and it is here that the point of measurement (H_a) is located.



- The throat (gray) of the flume accelerates the flow to a critical / super-critical state.
- The discharge section (light blue) slows the now energetic super-critical flow and allows it to transition into the downstream channel.
- While not all flumes have discharge sections (H / Montana flumes), the lack of this section means that the resistance to submergence is limited and that the downstream channel may be susceptible to

scour and erosion as the flow spills off the end of the flume.

Classes and Styles of Flumes

In the approximately 100 years since Dr. Parshall began his investigation on his Improved Venturi flume, two distinct classes (and numerous styles) of flumes have been developed.

Short-throated flumes

Short does not refer to the flume length but to the fact that flow is controlled in a very specific region of the flume to produce the level-to-flow relationship.

Openchannelflow manufactures the following styles of short-throated flumes:

- Parshall
- Montana (modified Parshall)
- USGS Portable (modified Parshall)
- HS / H / HL
- Trapezoidal
- Cutthroat

Short-throated flumes have a single, defined point of measurement and are empirical devices. Intermediate or non-standard sizes of short-throated flumes must be individually rated. Their use in **not recommended**.

Long-throated flumes

Long-throated flumes control flow in a throat that is long enough to

cause parallel flow lines in the section of flow control.

Openchannelflow manufactures the following styles of long-throated flumes:

- Palmer-Bowlus
- RBC

Long-throated flumes are somewhat more accurate than short-throated ones and can be computer designed (eliminating the need for costly laboratory rating).

Additionally, the point of measurement is **not fixed**, but can be taken anywhere upstream of the drawdown zone of the throat.